

On Power Domination of Generalized Petersen Graphs

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Abstract

The power dominating problem is a variation of the classical domination problem in graphs. Electricity company use phase measurement units (PMUs) to produce the measuring data of a system, and use these data to estimate states of the system. Because of the high cost of PMUs, minimizing the number of PMUs on a system is an important problem for electricity companies. This problem can be modeled by graphs as follows: Given an undirected graph $G = (V, E)$, find a set $S \subseteq V(G)$ with minimum cardinality such that all vertices in V are observed by some vertices in S . Herein, a vertex which is claimed to be a PMU observes itself and all its neighbors, and if an observed vertex has all but one of its neighbors observed, then the remaining neighbor becomes observed as well. The minimum size of S for graph G is called power dominating number and is denoted as $\gamma_p(G)$. This paper provides some tight upper bounds on power domination number of generalized Petersen graphs.

1 Introduction

The power domination problem arose in the context of monitoring electric power networks. A power network contains a set of nodes and a set of edges connecting the nodes. It also contains a set of generators which supply power, and a set of loads where the power is directed to [1]. It is important for electric power companies to

continuous monitoring the power systems and observing all the states, such as the voltage magnitude of loads and the current phase measurements at branches. A *Phase Measurement Unit* (PMU) is a measurement device placed on a node that has the ability to measure the voltage of the node and current phase of the edges that connected to the node, and to give warnings of system-wide failures. In order to monitor all the state variables of the network, they need to place PMUs at selected locations in the system. A system is said to be *observed* if all of the state variables of the system can be determined from a set of measurements (e.g., voltages, currents, and phase angle). More precisely, we have the following rules that tell us how vertex and edges can be measured:

1. A PMU on a vertex v measures the state variable for v and its incident edges and all adjacent vertices (these vertices and edges are said to be *observed*).
2. Any vertex that is incident to an observed edge is observed.
3. Any edge joining two observed vertices is observed.
4. If a vertex is incident to a total of $k > 1$ edges and if $k - 1$ of these edges are observed, then all k of these edges are observed.

One can increase the observed degree of the system by increasing the number of PMUs. However, because of the high cost of a PMU, it is economically beneficial to determine the optimal placement and the minimum number of PMUs

required while maintaining the ability to monitor (observe) the entire system. Hence the allocation of PMUs is an important factor to the observation degree.

Electric power networks can be interpreted as undirected graphs in a very natural fashion. Let $G = (V, E)$ be a graph representing an electric power system where a vertex represents an electrical node (a measuring point, a substation bus where transmission lines, loads, and generators are connected) and an edge represents a transmission line joining two electrical nodes. In order to observe the voltage and phase angle at some measuring point, we may place a PMU on it, and by using the propagation rules which are derived from the Ohm's and Kirchoff's laws for an electric circuit, we can observe measuring points indirectly as well. The problem of locating a smallest set of PMUs to monitor the entire system is a graph theory problem closely related to the well-known vertex covering and domination problems. Hence, this problem is not only interested in the power system industry but also interested as a new problem in graph theory. However, this type of domination is different from the standard domination problem since here the domination rules can be iterated.

2 Preliminaries

Let $v \in V(G)$. The *open neighborhood* of v , denoted as $N_G(v)$, is the set of vertices adjacent with v ; and the *closed neighborhood* of v , denoted as $N_G[v]$, is $N_G(v) \cup \{v\}$. For a set $S \subseteq V(G)$, the *open neighborhood* of S is defined as $N_G(S) = \bigcup_{v \in S} N_G(v) - S$ and the *closed neighborhood* of S is defined as $N_G[S] = N_G(S) \cup S$.

For a graph $G = (V, E)$, $S \subset V$ is a

dominating set of G if every vertex in $V - S$ has at least one neighbor in S . The *domination number* of G , denoted by $\gamma(G)$ is the minimum cardinality of a dominating set of G . The power domination may be considered as a variation of traditional domination. A set $S \subseteq V$ is a power dominating set (PDS) of G if every vertex not in S is observed by some vertices in S using the observation rules as follows [17]:

1. If a vertex is in PDS, it observed itself and all its adjacent vertices.
2. If a vertex has $k > 1$ adjacent vertices and $k - 1$ of these vertices are observed, then all k of these vertices are observed.

Notice that we simplify the rules to only consider the vertices since when all vertices are observed, all edges will be observed as well. The minimum cardinality among all PDS of G is the *power domination number* and is denoted as $\gamma_p(G)$. From now on, to be simplify, we use the term "domination" to substitute for "power domination".

The decision version of the power domination problem is as follow: Given an undirected graph $G = (V, E)$ and a nonnegative integer k , is there a subset $S \subseteq V$ with $|S| \leq k$ such that all vertices in V is observed according to the observation rules, if we placed a PMU on every $v \in S$?

This problem has been shown to be NP-complete even when restricted to bipartite graphs and chordal graphs [8], to planar cubic graphs [15], circle graphs [7][20], and to split graphs [12][20]. A variety of heuristic approaches to find power domination number for general graph or some certain graphs have been developed in the past two decades [7][10][11]. The problem is said to be *solved* for a graph if either a polynomial time algorithm of placing PMUs on the graph is given, or a formula of the power domination number for the graph is provided. The power domination problem has

efficient polynomial time algorithms for the classes of trees [8], graphs with bounded treewidth [7], block graphs[9][19], block-cactus graphs[9], interval[12], grids[14], honeycomb meshes[17] and circular-arc graphs [13]. Upper bounds on the power domination number are given for a connected graph with at least three vertices, for a connected claw-free cubic graph[21], for block graphs[19], for hypercube[3], and for generalized Petersen graphs [2][18]. Moreover, [3] and [21] independently showed that the power domination number in a connected graph with $n \geq 3$ vertices is no larger than $n/3$, and characterized the extremal graphs that attain the upper bound. Closed formulas for the power domination number are obtained for Mycielskian of the complete graph, the wheel, the n -fan and n -star [16], for Cartesian product of paths and cycles[2][6], for tensor and strong product of paths with paths[5], and for tensor product of paths with cycles[16].

3 Main Result

The generalized Petersen graph $P(m, k)$ for $m \geq 3$, and $m/2 > k \geq 1$, is defined as a graph, with vertex set $\{a_0, a_1, \dots, a_{m-1}\} \cup \{b_0, b_1, \dots, b_{m-1}\}$ and edge set $\{a_i b_i, a_i a_{i+1}, b_i b_{i+k} \mid i = 0, 1, \dots, m-1\}$ where the subscript sum is taken modulo m . An example of $P(7, 3)$ is illustrated in Figure 1.

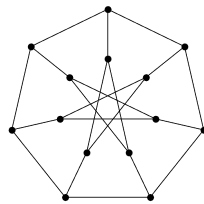


Figure 1: $P(7, 3)$

Theorem A (from [18]). Let $G = P(m, k)$ such that $m \geq 4$. Then $\gamma_p(G) \leq \min\{\lceil m/3 \rceil, k\}$.

Theorem B (from [18]). Let $G = P(m, k)$ such

that $m \equiv \pm 1 \pmod{k}$. Then $\gamma_p(G) \leq (m \mp 1)/k$.

Theorem C (from [18]). Let $G = P(m, k)$ such that $m = 3k$. Then

$$\gamma_p(G) = \begin{cases} 2, & \text{if } k = 2; \\ \lceil k/2 \rceil, & \text{if } k \neq 2. \end{cases}$$

Theorem D (from [2]). Let $G = P(m, k)$. Then $\gamma_p(G) \leq k$.

Theorem E (from [2]). Let $G = P(m, k)$ such that $m = 2k - 1$. Then $\gamma_p(G) = 2$.

Theorem 1. Let $G = P(m, k)$ such that $m = 4k$ and $k \geq 4$. Then $\gamma_p(G) \leq k - \lfloor k/3 \rfloor$.

Proof. $t = k - \lfloor k/3 \rfloor$. Consider placing the PMU on the set $S = \{a_0, a_3, \dots, a_{3t-3}\}$ such that $|S| = k - \lfloor k/3 \rfloor$, then by rule 1, $N[S] = \{a_{4k-1}, a_i \mid 0 \leq i \leq 3t-2\} \cup \{b_{3i} \mid 0 \leq i \leq t-1\}$ is power dominated by S . Since $N(a_1) = \{b_1, a_0, a_2\}$ and a_1, a_0 , and a_2 have been dominated, by rule 2, b_1 is also dominated by S . Repeating the same argument on $\{a_i \mid 2 \leq i \leq 3t-4 \text{ for } i \not\equiv 0 \pmod{3}\}$, then by rule 2, $\{b_i \mid 2 \leq i \leq 3t-4 \text{ for } i \not\equiv 0 \pmod{3}\}$ can be dominated by S . Now $\{a_i \mid 0 \leq i \leq 3t-2\}$, a_{4k-1} , and $\{b_i \mid 0 \leq i \leq 3t-3\}$ are already observed. Since $N(b_0) = \{b_k, a_0, b_{3k}\}$ and b_0, b_k , and a_0 have been dominated, by rule 2, b_{3k} is also dominated by S . Repeating the same argument on $\{b_i \mid 1 \leq i \leq k-3\}$, by rule 2, $\{b_i \mid 3k+1 \leq i \leq 4k-3\}$ can be dominated by S . Since $N(b_k) = \{b_0, b_{2k}, a_k\}$ and b_k, b_0 , and a_k have been dominated, by rule 2, b_{2k} is also dominated by S . Repeating the same argument on $\{b_i \mid k+1 \leq i \leq 2k-3\}$, by rule 2,

$\{b_i \mid 2k+1 \leq i \leq 3k-3\}$ can be dominated by S . Since $N(b_{2k}) = \{b_k, b_{3k}, a_{2k}\}$ and b_{2k}, b_k , and b_{3k} have been dominated, by rule 2, a_{2k} is also dominated by S . Repeating the same argument on $\{b_i \mid 2k+1 \leq i \leq 3k-3\}$ and $\{b_i \mid 3k+1 \leq i \leq 4k-3\}$, by rule 2, $\{a_i \mid 2k+1 \leq i \leq 3k-3\}$ and $\{a_i \mid 3k+1 \leq i \leq 4k-3\}$ can be dominated by S . Now the remaining vertex set which is not dominated by S is $\{b_{2k-1}, b_{2k-2}, b_{3k-1}, b_{3k-2}, b_{4k-1}, b_{4k-2}\} \cup \{a_{2k-1}, a_{3k-1}, a_{3k-2}, a_{4k-2}\}$. Since $N(a_{2k}) = \{a_{2k+1}, a_{2k-1}, b_{2k}\}$ and a_{2k}, a_{2k+1} , and b_{2k} have been dominated, by rule 2, a_{2k-1} is also dominated by S . Repeating the same argument on $\{a_{3k}, a_{3k-3}, a_{4k-3}\}$, by rule 2, $\{a_{3k-1}, a_{3k-2}, a_{4k-2}\}$ can be dominated by S . Since $N(a_{2k-1}) = \{a_{2k-2}, a_{2k}, b_{2k-1}\}$ and a_{2k-1}, a_{2k-2} , and a_{2k} have been dominated, by rule 2, b_{2k-1} is also dominated by S . Repeating the same argument on $\{a_{2k-2}, a_{3k-1}, a_{3k-2}, a_{4k-1}, a_{4k-2}\}$, by rule 2, $\{b_{2k-2}, b_{3k-1}, b_{3k-2}, b_{4k-1}, b_{4k-2}\}$ can be dominated by S . Now G is dominated by S . Hence S is a PDS of G . That is $\gamma_p(G) \leq k - \lfloor (k/3) \rfloor$. ■

Theorem 2. Let $G = P(m, k)$ such that $m = rk$ for some integer $r > 4$ and $k \geq 4$. Then $\gamma_p(G) \leq k+1 - \lfloor (k+1)/3 \rfloor$.

Proof. Let $t = k+1 - \lfloor (k+1)/3 \rfloor$. Consider placing the PMU on the set $S = \{a_0, a_3, \dots, a_{3t-3}\}$ such that $|S| = k+1 - \lfloor (k+1)/3 \rfloor$, then by rule 1, $N[S] = \{a_{m-1}, a_i \mid 0 \leq i \leq 3t-2\} \cup \{b_{3i} \mid 0 \leq i \leq t-1\}$

is power dominated by S . Since $N(a_1) = \{b_1, a_0, a_2\}$ and a_1, a_0 , and a_2 have been dominated, by rule 2, b_1 is also dominated by S . Repeating the same argument on $\{a_i \mid 2 \leq i \leq 3t-4 \text{ for } i \not\equiv 0 \pmod{3}\}$, then by rule 2, $\{b_i \mid 2 \leq i \leq 3t-4 \text{ for } i \not\equiv 0 \pmod{3}\}$ can be dominated by S . Now $\{a_i \mid 0 \leq i \leq 3t-2\}$, a_{m-1} , and $\{b_i \mid 0 \leq i \leq 3t-3\}$ are already observed. Since $k \geq 4$, $3t-2-k < 3t-3$. Since $N(b_{3t-2-k}) = \{b_{3t-2}, b_{3t-2-2k}, a_{3t-2-k}\}$ and $b_{3t-2-k}, b_{3t-2-2k}$, and a_{3t-2-k} have been dominated, by rule 2, b_{3t-2} is also dominated by S . Repeating the same argument on $\{b_i \mid 3t-1-2k \leq i \leq 3t-2\}$, by rule 2, $\{b_i \mid 3t-1-k < i \leq 3t-2+k\}$ can be dominated by S . Now $\{b_i \mid 0 \leq i \leq 3t-2+k\}$ is observed. Since $N(a_{3t-2}) = \{b_{3t-2}, a_{3t-3}, a_{3t-1}\}$ and a_{3t-2}, b_{3t-2} , and a_{3t-3} have been dominated, by rule 2, a_{3t-1} is also dominated by S . Repeating the same argument on $\{a_i \mid 3t-1 \leq i \leq 3t-2+k\}$, by rule 2, $\{a_i \mid 3t \leq i \leq 3t-1+k\}$ can be dominated by S . Now $\{a_i \mid 0 \leq i \leq 3t-1+k\}$ is already observed. Repeating the two processes above on $\{b_i \mid 3t-1 \leq i \leq m-k-1\}$ and $\{a_i \mid 3t-1+k \leq i \leq m-3\}$, by rule 2, G can be dominated by S . Hence S is a PDS of G . That is $\gamma_p(G) \leq k+1 - \lfloor (k+1)/3 \rfloor$. ■

By Theorem 2, we can conclude that if from a_0 to a_{2k} and from b_0 to b_{2k-1} are already dominated by a PMU set S , then $G = P(m, k)$ can be dominated by S , which implies S is a power dominating set of $G = P(m, k)$.

Theorem 3. Let $G = P(m, k)$ for $m > 3k$, and $k \geq 4$. Then $\gamma_p(G) \leq \lceil 2k/3 \rceil + 1$.

Proof. Let $t = \lceil 2k/3 \rceil + 1$. Consider placing the PMU on the set $S = \{a_0, a_3, \dots, a_{3t-3}\}$ such that $|S| = \lceil 2k/3 \rceil + 1$, then by rule 1, $N[S] = \{a_{m-1}, a_i \mid 0 \leq i \leq 3t-2\} \cup \{b_{3i} \mid 0 \leq i \leq t-1\}$ is power dominated by S . Since $N(a_1) = \{b_1, a_0, a_2\}$ and a_1, a_0 , and a_2 have been dominated, by rule 2, b_1 is also dominated by S . Repeating the same argument on $\{a_i \mid 2 \leq i \leq 3t-4 \text{ for } i \not\equiv 0 \pmod{3}\}$, then by rule 2, $\{b_i \mid 2 \leq i \leq 3t-4 \text{ for } i \not\equiv 0 \pmod{3}\}$ can be dominated by S . Now $\{a_i \mid 0 \leq i \leq 3t-2\}$, a_{m-1} , and $\{b_i \mid 0 \leq i \leq 3t-3\}$ are already observed. Since $3t-3 > 2k-1$ and b_{2k-1} is already dominated, then similar to Theorem 2, G can be dominated by S . Hence S is a PDS of G . That is $\gamma_p(G) \leq \lceil 2k/3 \rceil + 1$. ■

Algorithm 1.

Input: $G = P(m, k)$, $k \geq 4$ such that for any integer $r \geq 3$, $m \neq rk$, and $m \neq rk + k/2$.

Output: power dominating set S of G .

Method:

1. $S \leftarrow \{b_0\}$; temp = 0;
2. For $i \leftarrow 1$ to $\lceil 2k/3 \rceil + 1$ {
3. $S \leftarrow S \cup \{a_{ik - \lceil i/2 \rceil}\}$;
4. Check by the observation rule;
5. If G can be dominated by S {
6. temp = 1;
7. break;
8. }
9. }
10. If temp = 0
11. $S \leftarrow \{a_0, a_3, \dots, a_{3 \times \lceil 2k/3 \rceil}\}$
12. return S ;

End of Algorithm 1.

Theorem 4. Let $G = P(m, k)$, $k \geq 4$ such that $m \neq rk$ $r \geq 3$, and $m \neq rk + k/2$. Then Algorithm 1 produces a power dominating set of G

with cardinality no more than $\lceil 2k/3 \rceil + 1$ in $O(n^2)$ time.

proof. Algorithm 1 provides a set in two cases. The first case is checked by observation rule, so it is a power dominating set of G . The other case provides a set with cardinality $\lceil 2k/3 \rceil + 1$ which is proved in Theorem 3, so the second case is also a power dominating set of G . When a vertex is claimed to be a PMU, the algorithm will check that whether G is dominated S . The time complexity of running the observation rule is $O(n^2)$, and the maximum cardinality is $\lceil 2k/3 \rceil + 1$ which is proved in Theorem 3, so the time complexity of Algorithm 1 is $O(n^2)$. ■

4 Conclusion

This paper provides some tight upper bounds of power domination number on generalized Petersen graphs. The comparison of our result with known results is shown in Table 1:

Table 1: Results comparison

$G = P(m, k)$	
From [18]	Our result
$\gamma_p(G) \leq \min\{\lceil m/3 \rceil, k\}$	$\gamma_p(G) \leq \lceil 2k/3 \rceil + 1$
$G = P(rk, k)$	
From [18]	Our result
$\gamma_p(G) \leq \lceil k/2 \rceil$ for $r = 3$.	1. $\gamma_p(G) \leq k - \lfloor k/3 \rfloor$ for $r = 4$.
	2. $\gamma_p(G) \leq k + 1 - \lfloor (k+1)/3 \rfloor$ for $r > 4$.

For $P(m, k)$ where k is not a factor of m , this paper provided a sharper bound than the known result. When $P(rk, k)$, previous research only work on the case of $r = 3$, this paper discussed the case of $r \geq 4$.

Moreover, we give an $O(n^2)$ algorithm which produces a power dominating set of $P(m, k)$ with cardinality no more than $\lceil 2k/3 \rceil + 1$.

References

- [1] T.L. Baldwin, L. Mili, M.B. Boisen Jr., and R. Adapa, "Power System Observability With Minimal Phasor Measurement Placement," *IEEE Trans*, Vol. 8, pp. 707-715, 1993.
- [2] R. Barrera, and D. Ferrero, "Power Domination in Cylinders, Tori, and Generalized Petersen Graphs," *Networks*, Vol. 58, pp. 43-49, 2011.
- [3] D.J. Brueni, L.S. Heath, "The PMU placement problem," *SIAM J. Discrete Math*, Vol. 19, No. 3, pp. 744-761, 2005.
- [4] N. Dean, A. Ilic, I. Ramirez, J. Shen, K. Tian, "On the Power Dominating Sets of Hypercubes," *IEEE 14th International Conference on Computational Science and Engineering*, 2011.
- [5] P. Dorbec, M. Mollard, S. Klavzarz, S. Spacapan, "Power domination in product graphs," *SIAM J. Discrete Math*, Vol. 22, pp. 282-291, 2008.
- [6] M. Dorfling, and M. A. Henning, "A Note on Power Domination in Grid Graphs," *Discrete Appl Math*, Vol. 154, pp. 1023-1027, 2005.
- [7] J. Guo, R. Niedermeier, and D. Raible, "Improved Algorithm and Complexity Results for Power Domination in Graphs," *Lecture Notes in Computer Science Fundamentals of Computation Theory*, Vol. 3623, pp. 172-184, 2005.
- [8] T.W. Haynes, S.M. Hedetniemi, S.T. Hedetniemi, and M.A. Henning, "Domination in Graphs Applied to Electrical Power Networks," *SIAM J. Discrete Mathematics*, Vol. 15, pp. 519-529, 2002.
- [9] W.K. Hon, C.S. Liu, S.L. Peng, C.Y. Tang, "Power Domination on Block-cactus Graphs," *The 24th Workshop on Combinatorial Mathematics and Computation Theory*, 2007.
- [10] C.J. Hsu, *Power Domination on Bipartite Permutation Graphs of Maximum Degree 4*, Master Thesis, Department of Computer Science and Information Engineering, National Chi Nan University, 2008.
- [11] J. Kneis, D. Mölle, S. Richter, P. Rossmanith, "Parameterized power domination complexity," *Information Processing Letters*, Vol. 98, pp. 145-149, 2006.
- [12] C.S. Liao, and D.T. Lee, "Power Domination Problem in Graphs," *Lecture Notes in Computer Science Computing and Combinatorics*, Vol. 3595, pp. 818-828, 2005.
- [13] C.S. Liao, and D.T. Lee, "Power Domination in Circular-Arc Graphs," *Springer New York*, Vol. 61, 2011.
- [14] K.J. Pai, J.M. Chang, Y.L. Wang, "A simple algorithm for solving the power domination problem on grid graphs," In: *Proc. the 24th Workshop Combin. Math. and Comput. Theory*, pp. 256-260, 2007.
- [15] D. Raible, and H. Fernau, "Power Domination in $O^*(1.7548n)$ Using Reference Search Trees," *Lecture Notes in Computer Science*, Vol. 5369, pp. 136-147, 2008.
- [16] S. Varghese, and A. Vijayakumar, "Power Domination in Some Classes of Graphs," *EuroComb'11*, 2011.
- [17] Y.L. Wang, and G.S. Huang, *An Algorithm for Solving the Power Domination Problem on Honeycomb Meshes*, Master Thesis, Department of Computer Science and Information Engineering, National Chi Nan University, 2009.
- [18] G.J. Xu, and L.Y. Kang, "On the Power Domination Number of the Generalized Petersen Graphs," *J. Comb. Optim.* Vol. 22, No. 2, pp. 282-291, 2010.
- [19] G.J. Xu, L.Y. Kang, E.F. Shan, and M. Zhao, "Power Domination in Block Graphs," *Theoretical Computer Science*, Vol. 359, pp. 299-305, 2006.
- [20] M. Zhao, L. Y. Kang, "Power Domination in Planar Graphs with Small Diameter," *Journal of Shanghai University*, Vol. 11, pp. 278-222, 2007.
- [21] M. Zhao, L.Y. Kang, and G. J. Chang, "Power Domination in Graphs," *Discrete Math*, Vol. 306, pp. 1812-1816, 2006.